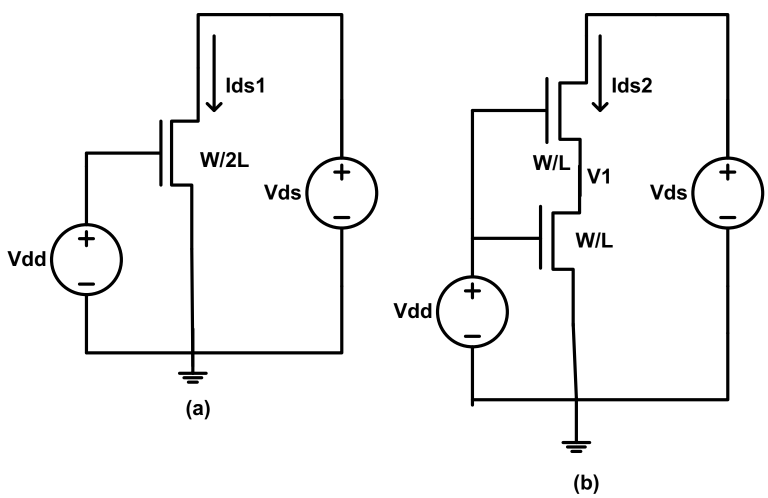
Homework 1

(Due 24th September in class)

Note: The first two problems are for practice purpose, will not be graded. You can work in a group of two students.

1) In the following two circuits show that $I_{DS1}=I_{DS2}$ when the transistors are in their linear region. First ignore the body effect. Assume the transistors to be long-channel device. If the body effect is considered, will I_{DS2} be equal to, greater that, or less than I_{DS1}? Please explain.



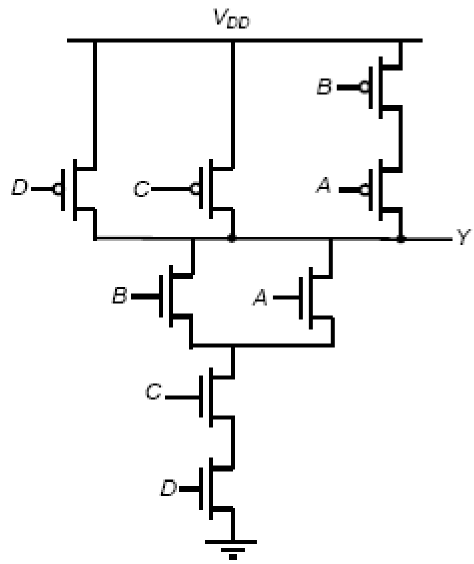
2) Compare I_{DS} versus V_{DS} for a NMOS transistor if $V_{GS} = 1.2V$, $V_{BS} = 0$, and $0<V_{DS}<1.2V$ for the following two case: (1) transistor is minimum size (W) and then transistor size is doubled (2W). Please use 65 nm technology parameters and use Cadence simulation.

3) Consider the following circuit

   a. What is the logic function implemented by the CMOS transistor network? Size the NMOS and PMOS devices so that the output resistance is the same as that of an inverter with an NMOS W/L = 4 and PMOS W/L = 8.

   b. What are the input patterns that give the worst case t_{pHL} and t_{pLH}. State clearly what the initial input patterns are and which input(s) has to make a transition in order to achieve this maximum propagation delay. Consider the effect of the capacitances at the internal nodes.

c. Verify part (b) with Cadence Spectra. Assume all transistors have minimum gate length and you are allowed to run the simulations in any technology node starting from 130 nm and below.



4) Consider the following two circuits

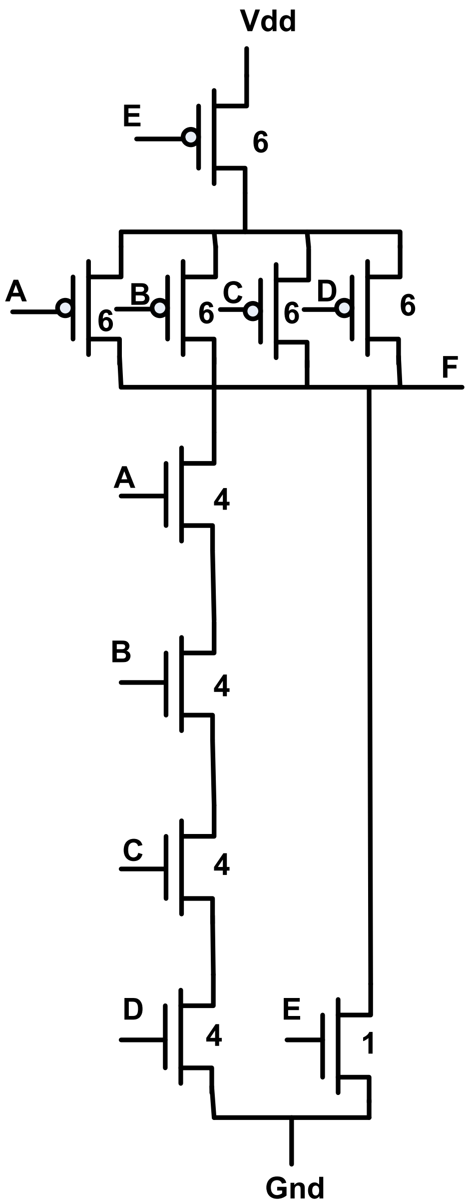
(a) Do the following two circuits implement the same logic function? If yes, what is that logic function? If no, give Boolean expressions for both circuits.

(b) Will these two circuits' output resistances always be equal to each other?

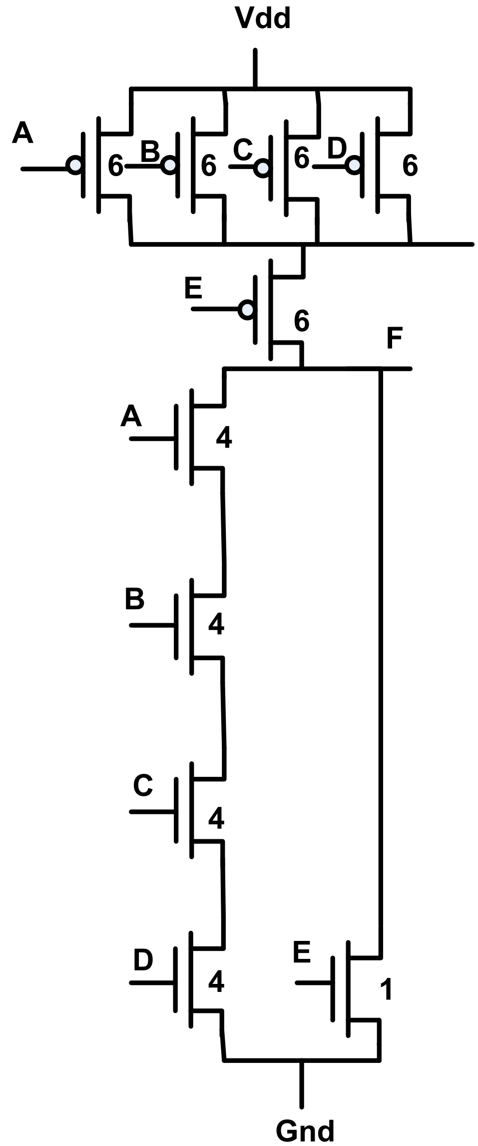
(c) Will these two circuits' rise and fall times always be equal to each other? Why or why not? (Demonstrate through Cadence simulation)

(d) What input patterns (*A*–*E*) give the lowest output resistance when the output is low?

(e) What input patterns (*A*–*E*) give the lowest output resistance when the output is high?

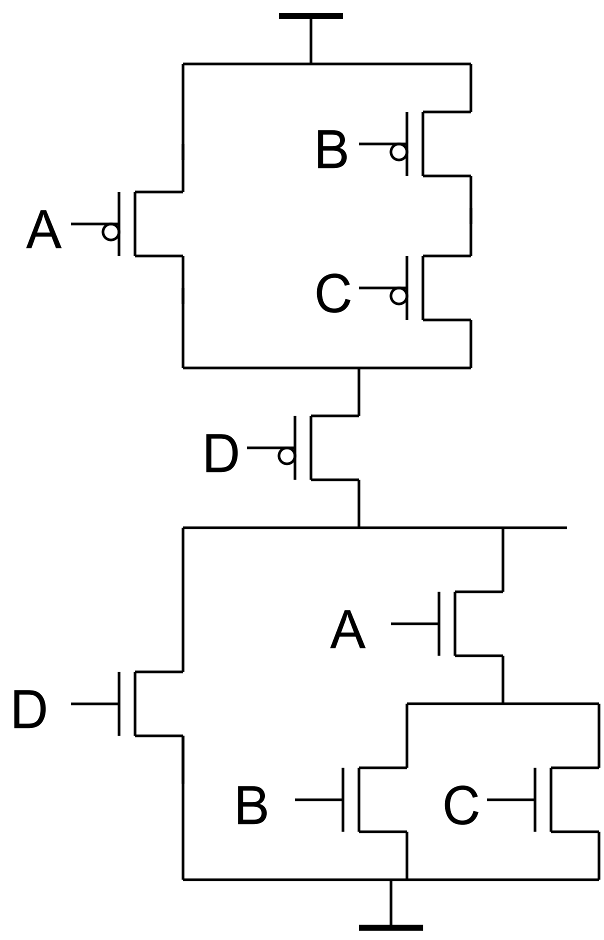


(a)



(b)

5) Determine the sizes of the transistors in the following figure, such that it has approximately the same t_{plh} and t_{phl} as an inverter with the following sizes: NMOS=W and PMOS=2W (First use hand calculation and then verify through Cadence simulation)



6) Implement the following function as a single stage CMOS gate and a pseudo NMOS gate

$$F = \overline{AB + CD}$$

Determine the size of the transistors for both CMOS and pseudo NMOS gates so that the high-to-low delays in both the cases are equal (use Cadence simulation)